

Table V.2: Hash Function Performance - 1 KB / 1MB Input, 2,000 Iterations

Algorithm	Input Size	Median Latency	Throughput (MB/s)	Notes
SHA-256	1 KB	Microseconds	~447 MB/s (1 MB scale)	Fastest overall throughput
SHA3-256	1 KB	Microseconds	~362 MB/s (1 MB scale)	Sponge-based; ~19% slower than SHA-256
SHAKE256	1 KB	Microseconds	~362 MB/s (1 MB scale)	Nearly identical to SHA3-256
SHA-256	1 MB	Low ms	~447 MB/s	Highest sustained throughput
SHA3-256	1 MB	Low ms	~362 MB/s	~19-20% below SHA-256
SHAKE256	1 MB	Low ms	~362 MB/s	Structurally identical to SHA3-256

Algorithm	Mean Time (s)	Std Dev	95% CI (s)	Skew	Shapiro p-value	Throughput (MB/s)
AES-128	0.000184	0.00003	(0.000180, 0.000188)	8.03	8.47E-27	5,436.92
AES-192	0.000213	0.000024	(0.000210, 0.000217)	5.14	2.84E-20	4,685.35
AES-256	0.000236	0.000023	(0.000233, 0.000240)	6.63	5.99E-22	4,229.64

Table V.3a: AES-CBC Encryption Performance File Size: 1 KB, 200 Iterations

Algorithm	Mean Time (s)	Std Dev	95% CI (s)	Skew	Shapiro p-value	Throughput (MB/s)
AES-128	0.008186	0.00073	(0.008084, 0.008288)	1.75	9.117E-13	1,221.54
AES-192	0.0084	0.001025	(0.008256, 0.008543)	1.39	5.201E-15	1,190.53
AES-256	0.009354	0.001293	(0.009173, 0.009535)	1.06	8.628E-11	1,069.08

Table V.3b: AES-CBC Encryption Performance File Size: 10 MB, 200 Iterations

Algorithm	Mean Time (s)	Std Dev	95% CI (s)	Skew	Shapiro p-value	Throughput (MB/s)
AES-128	0.081014	0.008416	(0.079838, 0.082190)	5.1	2.26E-23	1,234.35
AES-192	0.085283	0.00854	(0.084089, 0.086476)	2.07	1.23E-19	1,172.57
AES-256	0.085066	0.006045	(0.084221, 0.085911)	2.78	3.20E-20	1,175.56

Table V.3c: AES-CBC Encryption Performance File Size: 100 MB, 200 Iterations